



LABORATORY MANUAL

SC4172
IoT - TinyML

Lab Experiment #3

IMU-based Gesture Recognition

COLLEGE OF COMPUTING AND DATA SCIENCE
NANYANG TECHNOLOGICAL UNIVERSITY

1. **OBJECTIVES**

Enable custom gesture recognition via IMU sensors (Accelerometer and Gyro)

2. **LABORATORY**

This experiment is conducted at the **Software Laboratory 1 (Location: N4-01a-02)**.

3. **HARDWARE EQUIPMENT**

- Computer installed with Anaconda and Tools for TensorFlow ML.
- Arduino Nano 33 BLE Sense board
- Arduino IDE installed with TensorFlow Lite (TFLite) library

4. **REFERENCES**

- TinyML book by Pete Warden, Daniel Situnayake. 2019.
<https://www.oreilly.com/library/view/tinyml/9781492052036/>
- Nano33 BLE Sense Board. <https://docs.arduino.cc/hardware/nano-33-ble-sense/>
- Blog post on Gesture Recognition. <https://blog.tensorflow.org/2019/11/how-to-get-started-with-machine.html>
- SC4172 Lecture Notes

5. **INTRODUCTION**

In this lab, you will be introduced to a more in-depth tutorial you can use to train your own custom gesture recognition model for Arduino using TensorFlow. This lab reference closely to the blog post at <https://blog.tensorflow.org/2019/11/how-to-get-started-with-machine.html>

You will capture motion data from the Arduino Nano 33 BLE Sense board, import it into TensorFlow to train a model, and deploy the resulting classifier onto the board.

6. **EXERCISE**

The zip folder contains two Arduino programs:

- IMU Capture
 - program for you to generate the data (Punch and Flex) for training the model.
- IMU Classifier
 - programs to test the model you train.
 - Note that the IMU_Model.h file is the model I trained on Anaconda and converted for the Arduino Nano 33 BLE Sense board. This can be used to test your system setup.
 - After confirming that your setup is ok, you should capture your own data and train the model yourself.
- Please refer to the Gesture Recognition lecture note/video for discussion about this exercise.
- More detailed description about this exercise can be found at
 - <https://blog.tensorflow.org/2019/11/how-to-get-started-with-machine.html>
- The model training using Jupyter Notebook is available at the website, but it is based on Google's Colab

- https://colab.research.google.com/github/arduino/ArduinoTensorFlowLiteTutorials/blob/master/GestureToEmoji/arduino_tinyml_workshop.ipynb
- You should be able to use similar steps and do the training on your own laptop (which is how I did mine), and convert the model to the header file needed, based on what you had learnt in Lab 1 and 2. There are two more steps required to achieve this
 - Convert the trained model to the TFLite model and stored as TFLite for Microcontroller format.
 - Integrate the model with the application software and deploy it on the nano33 Sensor board.